

$$\mathbf{J}=\mathbf{E}\times\mathbf{H}$$

$$\frac{\mathbf{E}}{\mathbf{H}}$$

$$\frac{\mathbf{j}}{\mathbf{s}}$$

$$\mathbf{J}=\mathbf{S}+\mathbf{L}$$

$$(r,\phi,z)$$

$$e^{i\ell\phi}$$

$$L_z=\ell\hbar$$

$$\frac{L_z}{\hbar}$$

$$\frac{\partial E_x/\partial x}{\partial E_y/\partial y}$$

$$\nabla\cdot\mathbf{E}=\frac{\partial E_x}{\partial x}+\frac{\partial E_y}{\partial y}+\frac{\partial E_z}{\partial z}=0$$

$$\frac{E_z}{E_x},\frac{E_y}{s}=$$

$$\frac{\pm 1}{s}=\frac{1}{+1}$$

$$\mathbf{E}_{in}=\mathbf{E}_0(r,z)(\hat{x}+is\hat{y})e^{ikz}$$

$$\mathbf{E}_{in}\propto(\hat{x}+is\hat{y})=(\cos\phi\hat{\rho}-\sin\phi\hat{\phi})+is(\sin\phi\hat{\rho}+\cos\phi\hat{\phi})$$

$$\frac{\theta\hat{\phi}}{\hat{z}}$$

$$\frac{\hat{\rho}}{\hat{z}}$$

$$\frac{\hat{\rho}}{\hat{z}}$$

$$Projection onto z \propto \cos\phi + is \sin\phi$$

$$\cos\phi+is\sin\phi=e^{is\phi}$$

$$\frac{\theta}{\phi_p}$$

$$\frac{E_z}{\phi_p}$$

$$E_z\propto\int_0^{2\pi}(\cos\phi_p+is\sin\phi_p)e^{ik\rho\cos(\phi_p-\phi)}d\phi_p$$

$$\int_0^{2\pi}e^{is\phi_p}e^{ik\rho\cos(\phi_p-\phi)}d\phi_p=e^{is\phi}\int_0^{2\pi}e^{is\alpha}e^{ik\rho\cos\alpha}d\alpha$$

$$\frac{e^{is\phi}}{J_1(k\rho\sin\theta)}$$

$$\frac{\tilde{0}}{E_z(\rho,\phi,z)}$$

$$E_z=(\rho,\phi,z)=-iAe^{is\phi}I_1(\rho,z)$$

$$\frac{A}{I_1(\rho,z)}$$

$$\frac{NA}{n\sin\alpha}$$

$$I_1(\rho,z)=\int_0^\alpha\cos^{1/2}\theta\sin^2\theta P(\theta)J_1(k\rho\sin\theta)e^{ikz\cos\theta}d\theta$$

$$\frac{\theta}{\tilde{P}(\theta)}$$

$$\frac{J_1}{E_z}$$

$$\frac{e^{is\phi}}{\ell}\equiv$$

$$\frac{s}{s}=\frac{+1}{+1}$$

$$\frac{+1}{+1}\ell=$$

$$\frac{z}{z}$$

$$\left(\frac{\partial^2}{\partial r^2}+\frac{1}{r}\frac{\partial}{\partial r}+\frac{1}{r^2}\frac{\partial^2}{\partial \phi^2}+2ik\frac{\partial}{\partial z}\right)u(r,\phi,z)=0$$

$$\frac{\exp(i\ell\phi)}{2\pi}$$

$$\frac{\ell}{\ell} =$$